Willem Mirkovich

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Professional Work Experience

Senior Software Engineer
Moody's (*Acquired Cape Analytics*), Remote/NYC — May 2025 - Present

- **Technical Lead Across ML Platform Initiatives**: Served as technical lead across multiple ML platform initiatives, mentoring engineers, authoring design documents, delegating tasks, and coordinating handoffs with ML and product teams, including:
  - Classification model ingesting up to 50 home images per inference, enabling coverage across 30+ million locations
  - Integration of Digital Terrain Model (DTM) imagery into the model pipeline
  - Imagery embedding model with optimized tensor storage, reducing Google Cloud Storage costs by 95%
- **Re-architected Inference Service**: Re-architected the platform's primary ML orchestrator by extracting synchronous scikit-learn inference calls into a dedicated, horizontally scalable REST service. Enabled async invocation from the orchestrator, reducing end-to-end latency by 30% and handling 1.7k req/s on average.
- **Built Tool for Registration into Model Repository**: Built an internal MLOps tool for registering models to a centralized repository, handling validation, metadata management, and model packaging. Enabled seamless deployment to multiple targets including Nvidia Triton Inference Server and standalone FastAPI Docker containers, standardizing the model release process across the platform with 30+ models registered to date.

Senior Software Engineer
Cape Analytics, Remote — Feb 2023 - May 2025

- **Drove Performance Improvements, Dropping Client Latency by 40%**: Drove platform-wide performance improvements reducing client-facing latency by 40%, resolving SLA breaches across microservices. Added DataDog tracing and metrics to identify bottlenecks, then implemented targeted fixes:
  - Replaced Python schema validation with a Rust-based library, achieving a 99x average speedup
  - Rewrote a SQL query to use an intermediate mapping table, reducing query latency by 95%
  - Adopted Well-Known Binary (WKB) geometry encoding between services to reduce serialization overhead
- **Led Overhaul of Geospatial Imagery Sources**: Led overhaul of geospatial imagery sources across the platform to support new business and provider requirements, enabling near 100% coverage of Continental US. Onboarded multiple providers spanning raw GeoTIFF imagery and REST API services. Coordinated with ML teams to adapt foundational models to new imagery inputs, ensuring continued model accuracy across sources.
- **Deployed MCP Servers**: Deployed Model Context Protocol (MCP) servers across internal services, enabling developers to connect AI agents (Claude, ChatGPT) to internal data and query services via natural language prompts. Reduced the barrier for teams to integrate LLM tooling with existing infrastructure.

Software Engineer
Cape Analytics, Remote — Feb 2022 - Feb 2023

- **Integrated Neural Network Models**: Integrated deep learning PyTorch models processing geospatial imagery into the production platform, deploying to Nvidia Triton Inference Server on T4 GPUs. Optimized model serving for latency and GPU memory utilization across models ranging from MB to multi-GB in weight size.
- **Investigated and Fixed Critical Memory Leak**: Diagnosed and resolved a critical memory leak in the primary WMTS tile-serving service using py-spy and binary search across package versions, identifying a regression introduced in a rasterio release. Mitigated by reverting the dependency and filed a bug report with the open-source maintainer (rasterio/issues/2681).

Software Engineer
Anark Corporation, Boulder CO — Jun 2020 - Jan 2022

- **Migrated 3D CAD Model Rendering Codebase to TypeScript**: Migrated a 3D CAD model rendering codebase from JavaScript to TypeScript, improving type safety and reducing runtime bugs. Configured webpack to produce minified production bundles for browser deployment.

Skills Summary

- **Languages**: Python, SQL, TypeScript, JavaScript, Bash, Rust
- **ML / AI**: PyTorch, scikit-learn, Nvidia Triton Inference Server, webdataset
- **Geospatial / Computer Vision**: OpenCV, rasterio, Cloud Optimized GeoTIFFs
- **Web Frameworks**: FastAPI, Tornado, aiohttp
- **Infrastructure**: Docker, Kubernetes, Helm, Jenkins, DataDog, GCP, AWS, linkerd
- **Databases**: PostgreSQL, PostGIS, MongoDB, Google BigQuery

Education

- **University of Colorado Boulder, Engineering and Applied Science** — Boulder, CO
  Bachelor of Science - Computer Science; GPA: 3.9 — 2018 - May 2020
- **University of Washington Seattle, Pre-Engineering** — Seattle, WA
  Pre-Engineering - Computer Science; GPA: 3.7 — 2016 - June 2018